International Journal of Medical Informatics 139 (2020) 104143

Contents lists available at ScienceDirect

International Journal of Medical Informatics

journal homepage: www.elsevier.com/locate/ijmedinf

An Integrated Approach of Machine Learning and Systems Thinking for Waiting Time Prediction in an Emergency Department

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ARTICLE INFO

Keywords:
emergency departments
waiting time
machine learning
artificial intelligence
systems thinking

ABSTRACT

Objective: The objective of this study is to apply machine learning algorithms for *real-time* and *personalized* waiting time prediction in emergency departments. We also aim to introduce the concept of *systems thinking* to enhance the performance of the prediction models.
Methods: Four popular algorithms were applied: (i) stepwise multiple linear regression; (ii) artificial neural networks; (iii) support vector machines; and (iv) gradient boosting machines. A linear regression model served as a baseline model for comparison. We conducted computational experiments based on a dataset collected from an emergency department in Hong Kong. Model diagnostics were performed, and the results were cross-validated.
Results: All the four machine learning algorithms with the use of systems knowledge outperformed the baseline model. The stepwise multiple linear regression reduced the mean-square error by almost 15%. The other three algorithms had similar performances, reducing the mean-square error by approximately 20%. Reductions of 17 – 22% in mean-square error due to the utilization of systems knowledge were observed.
Discussion: The multi-dimensional stochasticity arising from the ED environment imposes a great challenge on waiting time prediction. The introduction of the concept of systems thinking led to significant enhancements of the models, suggesting that interdisciplinary efforts could potentially improve prediction performance.
Conclusion: Machine learning algorithms with the utilization of the systems knowledge could significantly improve the performance of waiting time prediction. Waiting time prediction for less urgent patients is more challenging.

1. INTRODUCTION

Emergency Department (ED) overcrowding is a long-standing issue confronting many countries and cities around the world [1–3]. This pervasive problem has drawn serious attention to the public because of its grievous consequences including putting public safety at risk, prolonged pain and suffering, and patients' dissatisfaction [4]. Waiting time has been one of the most common metrics for measuring ED efficiency. From the hospitals' perspective, a good approximation of waiting time helps EDs better manage their resources to anticipate potential overcrowding situations. From the patients' perspective, the dissemination of the waiting time to the public through the Internet or mobile applications is expected to assist non-urgent patients in informed decision-making – whether to visit the ED, continue to queue in the ED, or seek medical services at other healthcare facilities.

This research project aims to provide predictive models for patient waiting time estimation. Most EDs use the most recent observation of patient waiting time as the prediction; however, this approach does not anticipate future events, which must be considered in an ED environment as patients can wait for hours after arrival. The prediction of patient waiting time is highly complicated in practice due to the multi-dimensional stochasticity (e.g., patient arrivals, types of medical treatments and diagnostic tests required, and duration of treatments and tests). In the service industry, queueing models are popular for the estimation of waiting time; however, in an ED, many complicating factors (e.g., prioritization of patients according to their urgencies, the

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https://doi.org/10.1016/j.ijmedinf.2020.104143
Received 3 September 2019; Received in revised form 1 April 2020; Accepted 5 April 2020
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many different sequential and parallel service activities, and the work shifts of medical staff members) make the effectiveness of analytical modeling approaches unsatisfactory. Our work aims to utilize both *real-time* and *historical* operational data, powered by machine learning techniques, to achieve *personalized* and *more accurate* predictions. We also aim to demonstrate that data-driven approaches in combination with the concept of *systems thinking* [5] (i.e., the utilization of knowledge and underlying principles of how the system works) can help achieve a better predictive performance. The research questions are (i) whether machine learning approaches provided with ED operational data are effective in patient waiting time prediction; and (ii) whether the concept of systems thinking can enhance the performance of the machine learning algorithms.

The organization of this paper is as follows: We first present the material and methods in Section 2. Section 3 introduces the case study of our research. Section 4 describes the dataset used in our case study and how the dataset was processed. The results are presented in Section 5. Section 6 provides a discussion on our research findings, and Section 7 summarizes our conclusions and discusses future work.

2. MATERIAL and METHODS

2.1. Background and Related Work

Forecasting has been a particularly important topic in ED operations [6,7]. Linear regression was one of the most popular approaches to forecast waiting times or understand the relationships between waiting times and different factors [8,9]. In this study, a linear regression model was developed to serve as a baseline for comparison with our proposed approaches. There have been studies which developed more sophisticated data-driven methods for ED performance prediction. Hoot et al. [10] developed a discrete event simulation model to forecast crowdedness of ED in real time. Sun et al. [11] proposed quantile regression for real-time prediction of patient waiting time in an ED. Ang et al. [12] applied the Q-Lasso method with statistical learning and fluid model estimators to forecast ED patient waiting time. While existing research has a tendency to utilize large volumes of data to train prediction models, there is not much work on the combination of machine learning algorithms and the concept of systems thinking for waiting time prediction. Gartner and Padman [13] proposed an approach which incorporates a discrete-event simulation model into a machine learning algorithm. In this way, systems knowledge was utilized through the discrete-event simulation model. Our proposed methodology does not require a simulation tool to make predictions, but systems knowledge is used to derive useful information for training the machine learning models. Our work also offers a comparative study of the performances of widely adopted machine learning models.

2.2. Research Methodology

We conduct a case study of an emergency department in Hong Kong to examine the practicality and performance of our proposed integrated machine learning and systems thinking approach. Four machine learning models and two different sets of features are used to predict patient waiting times. Performance of the models will be tested through computational experiments.

2.2.1. Model Construction and Validation

In this study, four popular machine learning methods (which are commonly covered in machine learning textbooks for supervised learning for prediction of continuous outcomes, e.g., [14,15]) were applied and their performances were examined. These models include a baseline model and eight models trained by four approaches with two combinations of features.

Set **(a)** contained 11 features:

1 Patient's triage categories (three binary variables where each indicates if the patient is within the corresponding triage category; only urgent, semi-urgent and non-urgent patients are considered in our study);
2 Arrival time; and
3 Numbers of doctors within three hours of the patient's arrival (seven variables in total: three, two, and one hour (s) before the patient's arrival, upon the patient's arrival, and one, two, and three hour(s) after the patient's arrival).

Set **(b)** contained 18 features:

1 All features from (a);
2 Number of patients in queue for triage upon the patient's arrival;
3 Number of patients in queue for consultation in each category upon the patient's arrival (five categories in total) and
4 Number of patients in queue for departure upon the patient's arrival.

Feature set (b) can be considered as the additional information derived from the dataset with the utilization of systems knowledge. The baseline model LR(bl), which considers only triage category and arrival time, serves as our benchmark for comparison of models. The four modelling approaches used in this study were linear regression (LR), artificial neural networks (NN), support vector machines (SVM), and gradient boosting machines (GB). Hyperparameters of models NN, SVM, and GB were tuned using a grid search. A stratified five-fold cross-validation on the training set was applied for the grid search. Additional information on model hyperparameter tuning are presented in Appendix (A).

2.2.1.1. Baseline Model. Waiting time was regressed on the patient's triage category and arrival period. The time of arrival was stratified into two-hour intervals (from 12a.m. to 2a.m., from 2a.m. to 4a.m., etc.). Two-hour intervals were adopted to allow sufficient samples for training the models for Category 5 patients, whose observations were much fewer. The model was abbreviated as **LR(bl)**.

2.2.1.2. Linear Regression Models. Stepwise ordinary least squares regression models were fitted using the two feature sets and abbreviated as **LR(a)** and **LR(b)** respectively. First order interaction terms between the triage category and all other features were included in the model to account for the belief that these features would impose a different effect on patient waiting times of different triage categories. For example, the number of patients waiting for consultation is expected to have a significant impact on a newly arrived semi-urgent patient's waiting time, but not a critical patient's waiting time. Finally, to exclude insignificant predictors in the model, a backward stepwise model selection procedure was implemented with the minimization of the Akaike information criterion [16]. Model diagnostics including multicollinearity test by evaluating the variance inflation factor [17] and examination of heteroscedasticity (i.e., to test if the variance is a constant) using Breusch-Pagan test [18] were conducted.

2.2.1.3. Neural Networks Models. Two feedforward artificial neural networks models, **NN(a)** and **NN(b)**, were constructed. The NN models iteratively minimized mean-square error (MSE) of predicted values as a loss function through a stochastic gradient descent method, Adam, with a batch size of 512 [19]. All weights were initialized with Xavier initialization [20] and dropout was used to regularize the model [21]. The hyperparameters tuned by grid search included the number of hidden layers, the number of nodes in each hidden layer, choice of activation function and dropout rate. Learning curves were plotted to ensure that the models had converged.

2.2.1.4. Support Vector Machines. Support vector machines **SVM(a)** and **SVM(b)** were constructed with the two combinations of feature sets (a)

and (b). The hyperparameters tuned were the choice of the kernel function, parameters of the kernel function, penalty parameter of the error term, and epsilon in the soft margin loss function [22].

2.2.1.5. Gradient Boosting Machines. Stochastic gradient boosting machines **GB(a)** and **GB(b)** were constructed with the two feature sets. Weak decision trees were constructed for each boosting iteration [23]. To mitigate the effect of overfitting, regularization was performed by subsampling half of the training set in each iteration [24]. The learning rate, number of boosting iterations, maximum depth of each tree, maximum number of features considered and minimum number of samples to split a node were tuned by cross-validation.

2.2.2. Model Evaluation and Comparison

The performance of the models was compared by evaluating the test set prediction errors. In particular, the coefficient of determination R-squared, MSE, root-mean-square error (RMSE) were computed for each model. Predicted waiting times were plotted against actual values for each model to illustrate differences across models and feature sets. R and Python were used for data manipulation and analysis throughout the study.

3. CASE STUDY

We conduct computational experiments to examine the performance of the solution methodologies. Our computational study is based on a dataset collected from the ED of the Prince of Wales Hospital (PWH) in Hong Kong, one of the largest public hospitals in the city. The ED provides round-the-clock comprehensive emergency care to patients of all ages. A triage system is adopted in the ED; upon arrival, patients' clinical conditions are first assessed by a triage nurse, and their levels of urgency are classified into five categories: 1 (critical), 2 (emergency), 3 (urgent), 4 (semi-urgent), 5 (non-urgent). Priority of consultations and treatments is given to more urgent cases. The significant number of semi-urgent and non-urgent patient attendance (representing 70% of the overall attendance) to the ED demands a more accurate waiting time prediction methodology and offers a distinctive opportunity to conduct a case study of this ED. Our team has prior experience on the simulation and optimization of ED patient flows. Our previous key studies can be found in [25–28].

4. DATASET and DATA PROCESSING

Our analysis entailed six parts, including data retrieval, data preparation and feature engineering, outlier detection and splitting, model construction, model cross-validation and model evaluation (Fig. 1).

The computational experiments were conducted based on a dataset containing records of patient visits to the ED of PWH for December. The dataset contains information regarding each patient visit, including triage category, registration time, start times of service (triage and consultation) and departure time. In total, there were 12945 observations (i.e., patient visits) across the time span of 31 days. The work schedule of the doctors was used in the computational experiments as well.

4.1. Data Preparation and Feature Engineering

Missing data were observed in the dataset, and an imputation procedure was performed. Each missing timestamp was replaced by the timestamp of the previous event of the same patient visit plus an imputed time difference between the two events. The imputed time difference was set to the average time difference between the two events among all the patient visits in the same category whose arrival times were within three hours. If no observation was available for imputing a time difference, this time difference was estimated by mirroring the previous observation within the same triage category. The net time

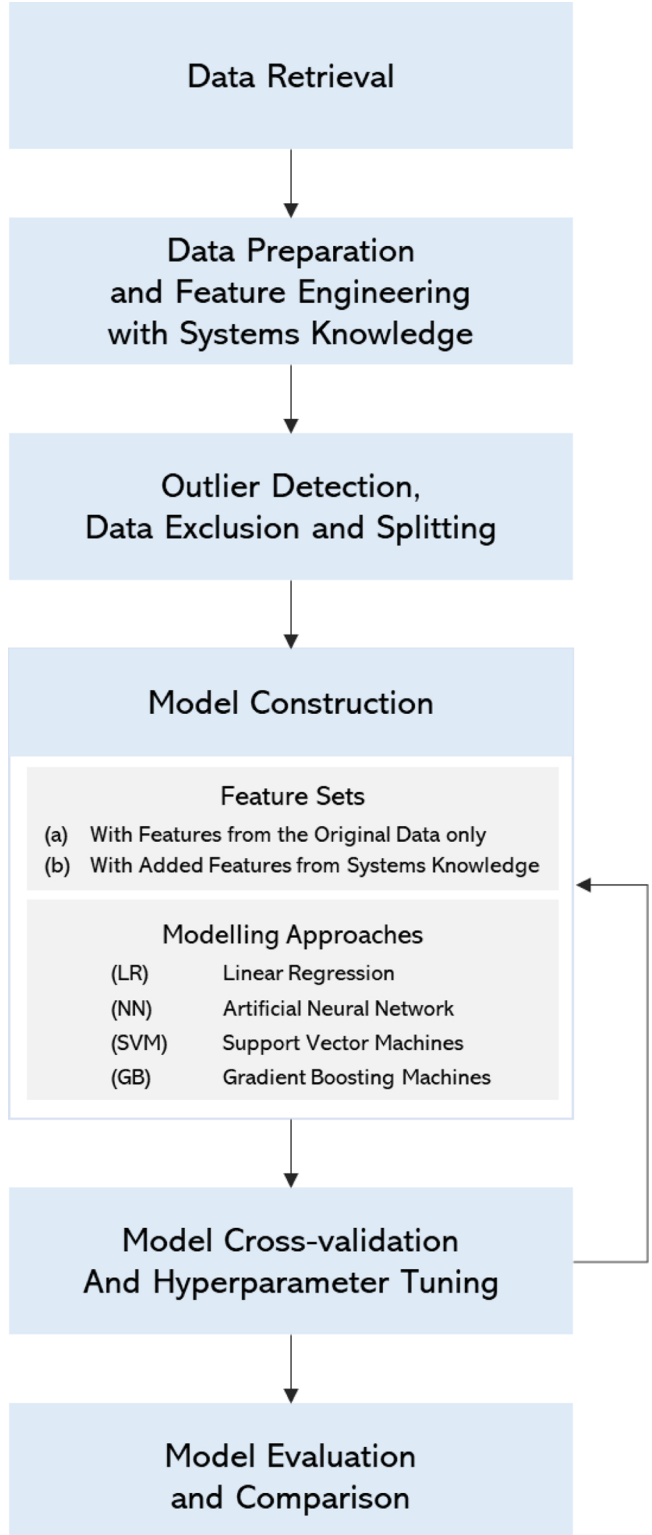

Fig. 1. Flowchart of the Data Handling and Analyses Process.
Note: Category 3 (urgent); Category 4 (semi-urgent); Category 5 (non-urgent).

from triage to consultation is considered to be the patient waiting time in this study. Features including triage category and patient registration time were used for waiting time prediction.

In addition to the data directly obtainable from the existing dataset, we also made use of the knowledge of the ED system. The flow of patients consists of a fixed sequence of activities, such as registration,

Table 1
Descriptive Statistics of Patients in the Training Set and Test Set.

	Training Set, Outliers Included	Training Set, Outliers Excluded	Test Set
Number of Observations	8,416	8,083	3,607
Category 3 (%)	2,321 (27.6%)	2,307 (28.5%)	995 (27.6%)
Category 4 (%)	5,975 (71.0%)	5,724 (70.8%)	2,561 (71.0%)
Category 5 (%)	120 (1.4%)	52 (0.6%)	51 (1.4%)
Age in years, mean (SD)	47.5 (25.6)	47.6 (25.7)	47.0 (25.8)
Gender, Male (%)	4,039 (48.0%)	3,870 (47.9%)	1,684 (46.7%)
Waiting time in minutes, mean (SD)			
Category 3	23.3 (40.0)	21.3 (23.3)	24.8 (45.7)
Category 4	112.6 (93.0)	104.8 (77.0)	107.9 (88.8)
Category 5	103.2 (101.0)	68.7 (55.5)	128.3 (87.0)
Number of doctors upon arrival, mean (SD)			
Category 3	5.9 (2.0)	5.9 (2.0)	6.0 (2.1)
Category 4	5.9 (2.0)	5.9 (2.0)	5.9 (2.0)
Category 5	5.7 (1.9)	5.3 (1.5)	6.0 (1.7)
Number of patients in queue for triage upon arrival, mean (SD)			
Category 3	2.0 (1.8)	2.0 (1.8)	2.0 (1.8)
Category 4	2.0 (1.9)	2.0 (1.9)	2.0 (1.9)
Category 5	2.1 (1.7)	2.0 (1.8)	2.3 (1.9)
Number of patients in queue for consultation upon arrival, mean (SD)			
Category 3	28.6 (16.1)	28.5 (16.0)	28.9 (16.0)
Category 4	27.0 (16.2)	26.0 (15.5)	26.2 (15.9)
Category 5	24.4 (15.2)	17.2 (12.0)	30.5 (16.2)
Number of patients in queue for departure upon arrival, mean (SD)			
Category 3	55.0 (17.6)	54.9 (17.5)	53.8 (17.8)
Category 4	52.7 (18.0)	52.0 (17.5)	52.0 (18.0)
Category 5	53.1 (15.8)	50.4 (14.4)	52.1 (17.3)

Note: Category 3 (urgent); Category 4 (semi-urgent); Category 5 (non-urgent).

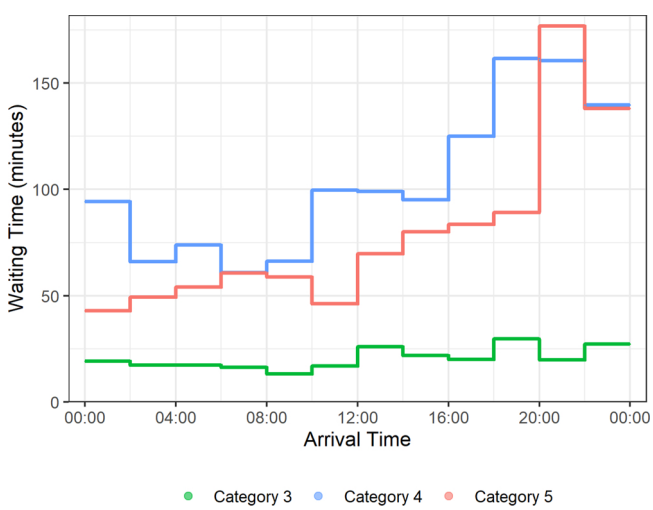

Fig. 2. Waiting Time by the Time Period and Triage Category Estimated by LR(bl).
Note: Category 3 (urgent); Category 4 (semi-urgent); Category 5 (non-urgent); LR (linear regression); NN (neural network); SVM (support vector machines); GB (gradient boosting machines); bl (baseline); a (feature set 1); b (feature set 2).

triage, consultation, and departure. The timestamps allow us to compute, upon a patient arrival, the total number of patients in the ED and queue lengths for triage and consultation. Records of patient visits missing the information about the triage category were filtered out because the information was critical in waiting time prediction. We also considered the number of physicians in the ED and their time effects. We calculated the numbers of doctors available in the ED at the time of the patient's arrival and one, two, and three hours before and after that (i.e., the numbers of doctors available at the seven time points). The derived information was mapped to the original observations as additional features.

4.2. Outlier Detection, Data Exclusion and Splitting

As most Categories 1 and 2 patients were usually seen in a timely manner, a waiting time prediction model may not be useful for them. Therefore, these observations were only used for calculating queue lengths but were excluded from model training and testing. All remaining observations were divided into a training set and a test set. For each triage category, 70% of observations belonged to the training set and the remaining belonged to the test set. As outliers were observed in the dataset, they were identified and removed from the training set to prevent the results from skewing towards the outliers. We adopted difference in fits (DFFITS), based on linear regression models, as the criteria for filtering. DFFITS is a technique to measure the influence of a point in a statistical regression model [29]. If the difference between the predicted value of the point and that calculated by excluding this observation in constructing the model is large, we consider this point as an outlier. Every observation in the training set was tested with DFFITS. If such difference was large, the observation was removed. Belsley et al. [30] suggested $2\sqrt{p/n}$ as a reasonable cutoff for DFFITS, where p is the number of parameters and n is the sample size. A stepwise linear regression model was constructed by using the same method as Model LR (b) to be introduced below, and observations with DFFITS greater than the suggested cutoff in magnitude were removed from the training set. On the other hand, the test set was held out from model training for evaluating and comparing model performances. Outliers were not removed from the test set in order to avoid underestimation of error and to report unbiased performance metrics. Finally, all features were normalized before model training such that all features had a zero mean and unit variance.

5. RESULTS

Among the 12,440 observations, 417 (3.4%) Category 1 or 2 patients were excluded. The training set contained 8,416 (70.0%) observations while the test set contained 3,607 (30.0%) observations. Cases with abnormally short or long waiting times were identified in the training set, and 333 (4.0%) were classified as outliers using

Table 2
Performance of the Models Evaluated with the Training and Test Sets

	LR(bl)	LR(a)	NN(a)	SVM(a)	GB(a)	LR(b)	NN(b)	SVM(b)	GB(b)
Train Set									
R-squared	0.377	0.352	0.388	0.360	0.400	0.528	0.599	0.568	0.699
MSE	3,607	3,750	3,541	3,707	3,476	2,732	2,323	2,501	1,740
RMSE									
Category 3	22.8	23.2	22.8	23.9	22.7	22.0	21.4	22.3	18.2
Category 4	69.8	71.1	69.1	70.6	68.4	60.5	55.6	57.7	48.1
Category 5	44.8	45.3	49.5	50.0	49.7	20.8	22.7	19.8	31.4
Test Set									
R-squared	0.268	0.249	0.274	0.238	0.271	0.374	0.410	0.401	0.429
MSE	5,618	5,764	5,575	5,851	5,598	4,803	4,528	4,597	4,383
RMSE									
Category 3	45.8	45.9	45.7	46.7	45.6	44.6	44.8	45.1	44.3
Category 4	82.9	84.2	82.6	84.7	82.9	76.4	74.0	74.5	72.5
Category 5	106.5	100.9	103.4	105.1	101.9	89.1	78.7	81.2	86.5
RMSE / SD									
Category 3	0.499	0.508	0.499	0.523	0.497	0.481	0.468	0.488	0.398
Category 4	0.786	0.801	0.778	0.795	0.77	0.681	0.626	0.65	0.542
Category 5	0.515	0.521	0.569	0.575	0.571	0.239	0.261	0.228	0.361
% Difference between (b) and (a)									
Train Set									
R-squared	NA	NA	NA	NA	NA	50.00%	54.38%	57.78%	74.75%
MSE	NA	NA	NA	NA	NA	−27.15%	−34.40%	−32.53%	−49.94%
RMSE									
Category 3	NA	NA	NA	NA	NA	−5.17%	−6.14%	−6.69%	−19.82%
Category 4	NA	NA	NA	NA	NA	−14.91%	−19.54%	−18.27%	−29.68%
Category 5	NA	NA	NA	NA	NA	−54.08%	−54.14%	−60.40%	−36.82%
Test Set									
R-squared	NA	NA	NA	NA	NA	50.20%	49.64%	68.49%	58.30%
MSE	NA	NA	NA	NA	NA	−16.67%	−18.78%	−21.43%	−21.70%
RMSE									
Category 3	NA	NA	NA	NA	NA	−2.83%	−1.97%	−3.43%	−2.85%
Category 4	NA	NA	NA	NA	NA	−9.26%	−10.41%	−12.04%	−12.55%
Category 5	NA	NA	NA	NA	NA	−11.69%	−23.89%	−22.74%	−15.11%
RMSE / SD									
Category 3	NA	NA	NA	NA	NA	−5.31%	−6.21%	−6.69%	−19.92%
Category 4	NA	NA	NA	NA	NA	−14.98%	−19.54%	−18.24%	−29.61%
Category 5	NA	NA	NA	NA	NA	−54.13%	−54.13%	−60.35%	−36.78%
% Difference between the Model and LR(bl)									
Train Set									
R-squared	NA	−6.63%	2.92%	−4.51%	6.10%	40.05%	58.89%	50.66%	85.41%
MSE	NA	3.96%	−1.83%	2.77%	−3.63%	−24.26%	−35.60%	−30.66%	−51.76%
RMSE									
Category 3	NA	1.75%	0.00%	4.82%	−0.44%	−3.51%	−6.14%	−2.19%	−20.18%
Category 4	NA	1.86%	−1.00%	1.15%	−2.01%	−13.32%	−20.34%	−17.34%	−31.09%
Category 5	NA	1.12%	10.49%	11.61%	10.94%	−53.57%	−49.33%	−55.80%	−29.91%
Test Set									
R-squared	NA	−7.09%	2.24%	−11.19%	1.12%	39.55%	52.99%	49.63%	60.07%
MSE	NA	2.60%	−0.77%	4.15%	−0.36%	−14.51%	−19.40%	−18.17%	−21.98%
RMSE									
Category 3	NA	0.22%	−0.22%	1.97%	−0.44%	−2.62%	−2.18%	−1.53%	−3.28%
Category 4	NA	1.57%	−0.36%	2.17%	0.00%	−7.84%	−10.74%	−10.13%	−12.55%
Category 5	NA	−5.26%	−2.91%	−1.31%	−4.32%	−16.34%	−26.10%	−23.76%	−18.78%
RMSE / SD									
Category 3	NA	1.80%	0.00%	4.81%	−0.40%	−3.61%	−6.21%	−2.20%	−20.24%
Category 4	NA	1.91%	−1.02%	1.15%	−2.04%	−13.36%	−20.36%	−17.30%	−31.04%
Category 5	NA	1.17%	10.49%	11.65%	10.87%	−53.59%	−49.32%	−55.73%	−29.90%

Note: Category 3 (urgent); Category 4 (semi-urgent); Category 5 (non-urgent); MSE (mean squared error); RMSE (root mean square error); LR (linear regression); NN (neural networks); SVM (support vector machines); GB (gradient boosting machines); bl (baseline); a (feature set 1); b (feature set 2).

DFFITS. Thus, 8,083 (96.0%) observations were included in the final training set. Descriptive statistics of the different categories of patients were presented in Table 1.

All the nine models were fitted. Results of LR(bl) can be interpreted as the predictions based on only two sets of categorical variables regarding the triage category and arrival time period (see Fig. 2). For model diagnostics of LR(a) and LR(b), there was no multicollinearity problem among the independent variables. However, the model failed the Breusch-Pagan test at a 5% significance level, implying that error terms of the model had non-constant variance. This violates the assumptions of ordinary least squares linear regression, leading to a loss in efficiency in estimating model parameters [18]. A summary of variables included in LR(b) as well as the detailed settings and hyperparameters for NN, SVM, and GB models after cross-validation are provided in Appendices (A) and (B). Performance metrics resulting from the nine models are presented in Table 2. All the models using feature set (a) were similar to the baseline model LR(bl) in terms of MSE. However, the performance of the models were significantly improved after the knowledge of the ED system was utilized, i.e., after deriving the queue lengths from the primary dataset and including them as additional features. LR(b) was able to reduce around fifteen percent of the MSE from the baseline model. On the other hand, the

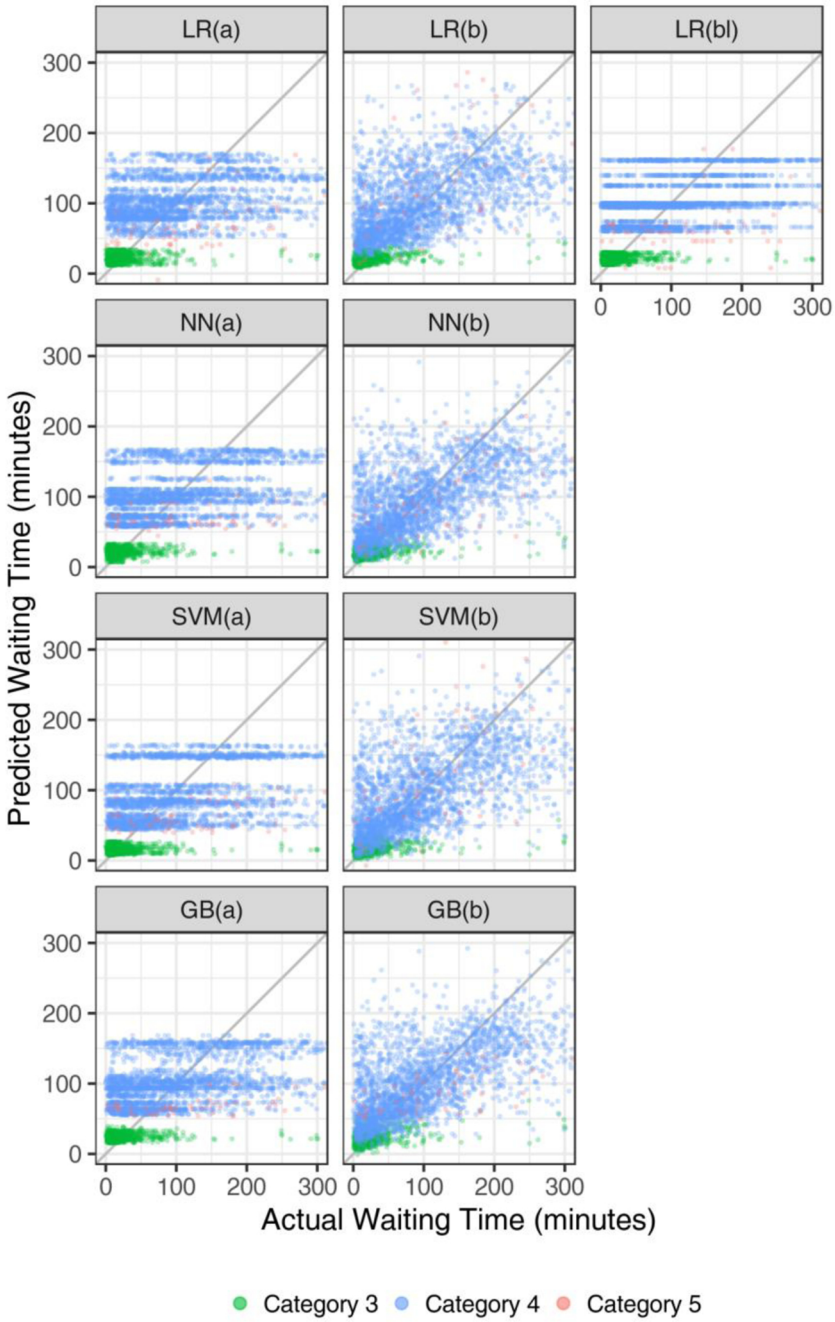

Fig. 3. Predicted Waiting Time against Actual Waiting Time for Different Models.
Each dot represents an observation in the test set. Triage categories of the patient are represented by different colours. The diagonal line in each plot represents the line where predicted waiting time equals to actual waiting time. A good model is expected to produce a plot where the dots are close to the diagonal line.
Note: Category 3 (urgent); Category 4 (semi-urgent); MSE (mean squared error); RMSE (root mean square error); LR (linear regression); NN (neural network); SVM (support vector machines); GB (gradient boosting machines); bl (baseline); a (feature set 1); b (feature set 2).

three machine learning models NN(b), SVM(b) and GB(b) outperformed LR(b) with no significant difference in performance among the three. They were able to reduce around twenty percent of MSE from LR(bl). Among all models, GB(b) had the greatest R-squared and the least MSE. The errors of the predicted waiting time in the test set for each model are presented in Fig. 3. All models with feature set (a) appears to have difficulty in identifying crucial features that impact waiting time. Their waiting time predictions had high bias and covered only a relatively small range of values, which were similar to LR(bl).

From the values of R-squared and MSE presented in Table 2, with the full set of features, NN(b), SVM(b) and GB(b) were superior to LR (b). Existing studies have reported that more sophisticated machine learning models outperform LR models in various applications [31,32], mainly due to the fact that many machine learning approaches can capture complex non-linear effects [31] while LR models only establish linear relationships between predictors and the dependent variable. Although such shortcoming of LR models may be overcome by performing variable transformations and introducing interaction and higher-order terms to the model, machine learning models have been observed to have better capabilities of capturing non-linear relationships. Our findings are consistent with existing studies that machine learning models outperform LR models on volatile datasets.

While machine learning models show better performance over the LR models, our study demonstrates that the knowledge of the ED system (i.e., the ED processes and operations) could improve the performance of the waiting time prediction, as compared with using machine learning approaches alone. Systems knowledge could be utilized to derive new features, which could potentially train more accurate machine learning models. This also suggests that there is a need for interdisciplinary research where data scientists can develop algorithms while domain experts can share their knowledge and provide feedback.

Another observation is that all the models studied had an inferior

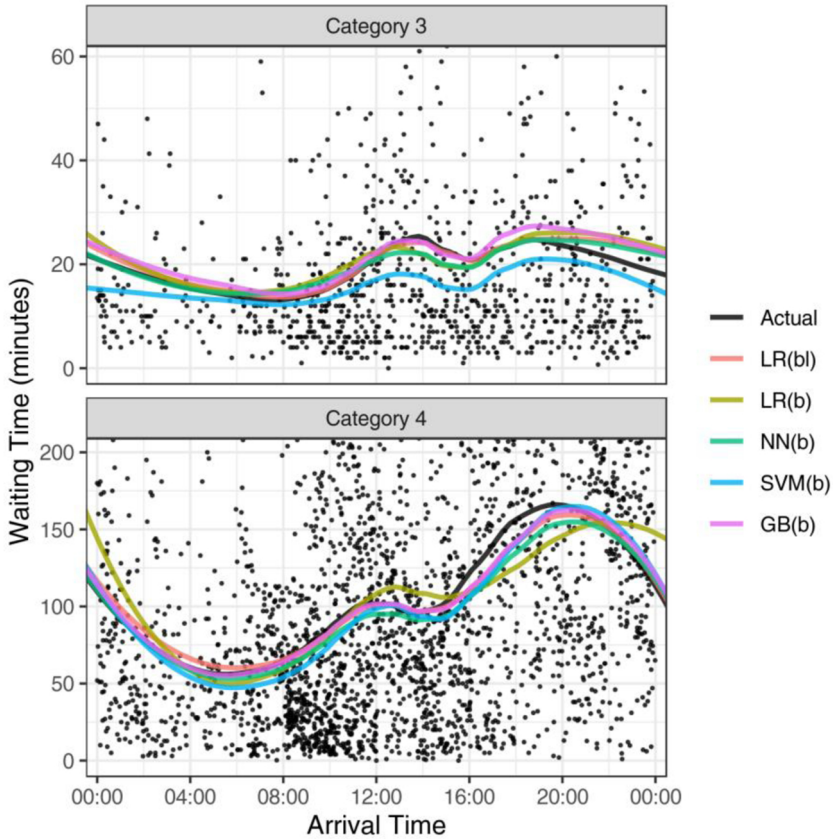

Fig. 4. Waiting Time against Time of Arrival for Categories 3 and 4 Patients.
Each dot represents an observation in the test set. The actual waiting time and predicted waiting times were smoothed across different observations and illustrated as curves.

Table 3
Comparison of Actual Waiting Time of Three Selected Semi-Urgent (Category 4) Patients.

Patient	Actual Waiting Time, in minutes	Number of Patients in Queue for		
		Triage	**Consultation**	**Departure**
A	9	1	14	45
B	75	2	14	48
C	119	2	14	47

In this example, there were three semi-urgent patients (A, B and C) arriving in a 10-minute interval. Upon each of their arrivals, the number of patients in queue (for triage, consultation, and departure) was similar to the others. However, their actual waiting times had significant differences.

performance for predicting waiting times of Category 4 patients, when compared to other triage categories, as shown by the rows of MSE/SD in Table 2. This can be explained by the higher variation in waiting time of Category 4 patients, as shown in Fig. 4. This high variation was due to the fact that Category 4 patients have a lower priority for receiving medical services and thus their waiting times are more sensitive. There were also cases that patients who were originally classified as Category 4 might need immediate medical attention due to the presence of certain symptoms. For other categories, with the same set of features for training, the performance of LR models was comparable to that of the machine learning models or even better. It indicates the importance of the quantity of training data required for the development of machine learning models. Our training set contained around 6,000 Category 4 patient visits and less than 100 Category 5 patient visits. With fewer training samples for Category 5 patient visits, the machine learning models appeared to be less effective in fitting the larger number of model parameters considered.

While machine learning models were observed to perform better in prediction, its “black box” nature does not provide a straightforward way to draw insights and interpret the relative effect of each variable in the prediction process, which has been widely discussed [33]. On the other hand, LR models could help interpret the results and understand the relative importance of the prediction features. For example, according to LR(b), if the number of Category 4 patients in queue for consultation increases by one, the expected waiting time for a newly arrived Categories 3, 4, and 5 patients would increase by 0.18, 2.75 and 5.77 minutes respectively. The coefficient increases with the triage category because the ED prioritizes the patients according to the triage category; the queue length has a more significant impact on the waiting times of patients with higher triage category (i.e., less urgent patients).

6. DISCUSSION

The stochastic nature of the many processes in an ED (e.g., patient arrivals, their required treatments, and treatment times) imposes a great challenge on the prediction of waiting time. Table 3 highlights an example where three Category 4 patients who arrived at the ED within a 10-minute interval, but it turned out that they had drastically

different waiting times. Critically-ill patients may be sent to the ED any time of day and physicians must provide immediate care to them upon their arrival. While a non-urgent patient is waiting at the ED, the presence of new patients who require urgent medical attention can drastically increase the waiting time. Also, the actual practice of operations may vary due to some special circumstances (e.g., physicians speeded up). Therefore, outlier detection was a crucial step in this study.

The ultimate goal of our study was to provide a more accurate waiting time prediction methodology. This information would be essential for EDs to make more responsive and proactive actions if long waiting time in the future can be anticipated. For example, the Hospital Authority of Hong Kong has a service target which requires 90% of Category 3 patients to be treated within 30 minutes. Upon a Category 3 patient's arrival, if the predicted waiting time is longer than 30 minutes, the hospital may ask the doctor-on-call to be on duty or deploy doctors from other divisions to the consultation rooms for more proactive management. The provision of such information to the patients could also improve patients' satisfaction in the ED [34]. Waiting times less than anticipated could positively impact overall satisfaction [34]. Thus, there would be important research questions following our study. Shall an ED provide an overestimated waiting time to improve patients' satisfaction? How would the patients' behavior change if long waiting time is announced to the patients? Research efforts and ethical concerns will be required to answer these questions. Nevertheless, a more accurate waiting time prediction would serve as a foundation to develop workable solutions related to these questions.

7. CONCLUSION AND FUTURE WORK

Emergency department (ED) overcrowding has been an important global issue for decades. The estimation of patient waiting time is crucial for responsive resource allocation and patients' informed decision to attend or stay at an ED. Accurate waiting time estimation is also expected to improve staffing decisions, leading to enhanced patient flow and satisfaction [35]. Our study aimed to apply machine learning algorithms, trained by the operational data captured at an ED, for real-time and personalized waiting time prediction. Four popular methods – stepwise multiple linear regression, artificial neural networks, support vector machines, and gradient boosting machines – were compared with the benchmark model, linear regression (LR). A major contribution of our work was the utilization of the knowledge of the ED system to derive further features for constructing more effective machine learning models. Computational experiments, based on a dataset of an ED in Hong Kong, were conducted to evaluate the performance of our approaches.

Our study shows that machine learning models were generally superior to LR models, with around 15 - 20% reduction in mean-square error. Reductions of 17% - 22 % in mean-square error due to the utilization of systems knowledge were observed. Our findings suggest that the knowledge of the system plays an important role in enhancing the performance of the prediction models, indicating that there is a need for interdisciplinary research where data scientists and domain experts work together.

There is future work following this research study. There is additional information that might be useful for improving the performance of the prediction algorithms. The dataset we utilized in this study was only operational data regarding the ED internal activities. Since delay in ED boarding to inpatient wards is one of the causes of ED overcrowding, information regarding inpatient activities may also be a major factor for waiting time prediction. Another future research direction is to investigate the use of real-time waiting time prediction information. How hospital management can utilize this piece of information for operational decision-making, e.g., resource allocation and patient diversion, would be an interesting and important research question.

Ethics approval

Ethics approval was granted by CUHK-NTEC CREC in Hong Kong (CREC Ref. No: 2018.324).

Clinical trial Registration

Not applicable.

FUNDING

This research is supported by Health and Medical Research Fund, Food and Health Bureau, the Hong Kong SAR Government [Project No: 14151771].

Contributors

YHK, JML, CAG, HM, AMS, and KKT initiated and designed the study. YHK formulated the research questions, designed the computing algorithms and computational experiments, and finalized the paper. NBC developed the computational programmes, conducted the data analysis, and drafted the initial manuscript. YHK, JML, HM, and AMS were involved with the conception and overall design of the methodologies. CAG and KKT provided knowledge of hospital operations and contributed to the research findings from a healthcare perspective.

Summary Points

What was already known

- Numerous studies conducted in different countries have reported emergency department overcrowding for decades, but until now this problem is still a serious global issue.
- The estimation of patient waiting time is crucial for responsive resource allocation at emergency departments.
- The current practice of patient waiting time estimation typically uses the average of current observations or linear regression models as prediction, but these estimations are not personalized and do not capture information on the workload in the system.

What this study has added

- Our study shows that machine learning models are generally superior to linear regression models.
- The knowledge of the system is shown to play an important role in enhancing the performance of the prediction models.

The authors declare no competing interests.

We thank the staff of the Emergency Department of the Prince of Wales Hospital.

Appendix B. (B). Summary of Variables Included in LR(b)

Variables	Coefficient (95% CI)	p-value
Intercept term	16.24 (8.59, 23.90)	< 0.001
Category 4	41.5 (31.87, 51.13)	< 0.001
Category 5	74.54 (3.70, 145.39)	0.039
Time of day (in hours)	0.31 (-0.08, 0.69)	0.120
Time of day (in hours) and Category 4	2.53 (2.05, 3.00)	< 0.001
Number of doctors upon arrival	−0.79 (-1.52, -0.05)	0.036
Number of doctors upon arrival, given Category 5	10.56 (-0.17, 21.28)	0.054
Number of doctors 1 hour after arrival	−0.85 (-1.95, 0.25)	0.131
Number of doctors 1 hour after arrival, given Category 4	−3.96 (-5.30, -2.61)	< 0.001
Number of doctors 1 hour after arrival, given Category 5	−16.89 (-26.41, -7.36)	0.001
Number of doctors 2 hours before arrival, given Category 4	−1.37 (-2.14, -0.61)	< 0.001
Number of doctors 2 hours after arrival, given Category 4	−3.18 (-3.99, -2.37)	< 0.001
Number of patients in queue for triage	0.42 (-0.23, 1.06)	0.202
Number of category 1 patients in queue for consultation, given Category 5	0.00 (0.00, 0.00)	0.015
Number of category 2 patients in queue for consultation, given Category 4	−8.47 (-14.44, -2.50)	0.005
Number of category 3 patients in queue for consultation	2.18 (1.57, 2.80)	< 0.001
Number of category 4 patients in queue for consultation	0.18 (0.03, 0.34)	0.023
Number of category 4 patients in queue for consultation, given Category 4	2.57 (2.38, 2.75)	< 0.001
Number of category 4 patients in queue for consultation, given Category 5	5.59 (3.85, 7.33)	< 0.001
Number of category 5 patients in queue for consultation, given Category 5	−13.97 (-33.04, 5.10)	0.151
Number of patients in queue for departure, given Category 5	−1.03 (-2.18, 0.12)	0.080

Category 3 patients were taken as the reference group.

Appendix C. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.ijmedinf.2020.104143.

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